DA3452 – Speech and Language Technology

Module Test 1

Representing Language Computationally

Duration:	60 minutes	**Maximum marks:**	45
Questions:	30	**Negative marking:**	None

Instructions. Choose the single best answer for each question. For every question, fill the corresponding bubble on the OMR sheet *and* write the selected letter in the final-answer space. The filled bubble is the primary response; the handwritten letter will be used only to resolve an unclear bubble. If both are clear but disagree, the filled bubble will be evaluated. Questions 1–15 carry 1 mark each; Questions 16–30 carry 2 marks each.

Useful reminder (UTF-8 byte ranges). U+0000–U+007F uses 1 byte; U+0080–U+07FF uses 2 bytes; U+0800–U+FFFF uses 3 bytes; and U+10000–U+10FFFF uses 4 bytes.

Useful conventions. Unless otherwise stated, use

$$\mathrm{tf}(t,d) = \begin{cases} 1 + \log_{10} c(t,d), & c(t,d) > 0, \\ 0, & c(t,d) = 0, \end{cases} \qquad \mathrm{idf}(t) = \log_{10}\left(\frac{N}{\mathrm{df}_t}\right), \qquad \mathrm{tfidf}(t,d) = \mathrm{tf}(t,d)\,\mathrm{idf}(t).$$

Section A – 1-mark questions

1. Which statement best captures why a word is not a universal computational unit for language processing?
   (A) Every written language has spaces, but computers cannot reliably detect them.
   (B) Word boundaries and even the notion of a word can depend on language, writing system, and task.
   (C) Words cannot be represented by Unicode.
   (D) Every word must contain more than one morpheme.

2. As a corpus becomes much larger, its vocabulary usually continues to grow, but more slowly than the number of running words. What is the most important computational consequence emphasized in the reading?
   (A) A sufficiently large whole-word vocabulary eventually contains every possible future word.
   (B) Unknown words disappear once the corpus is large enough.
   (C) Whole-word vocabularies remain open-ended, motivating representations that can compose unseen words from smaller units.
   (D) The number of word types eventually becomes independent of corpus size.

3. A corpus contains 1,000 running word occurrences but only 420 distinct word forms. In the terminology of Jurafsky–Martin, which statement is correct?
   (A) It has 1,000 word types and 420 word instances.
   (B) It has 420 word types and 1,000 word instances.
   (C) It has 420 tokens and therefore exactly 420 morphemes.
   (D) The type/instance distinction applies only after BPE tokenization.

4. Consider two Python strings: one contains precomposed é (U+00E9), and the other contains `e` followed by COMBINING ACUTE ACCENT (U+0065 U+0301). Before NFC normalization, which statement is correct?
   (A) They may look identical but have different code-point sequences and different lengths under `len()`.
   (B) They must compare equal because they look identical.
   (C) They have different glyphs and therefore cannot be canonically equivalent.
   (D) UTF-8 converts both to the same bytes automatically.

5. Why is every ASCII text file also a valid UTF-8 text file for the ASCII characters it contains?
   (A) UTF-8 stores every character in four bytes, including ASCII.
   (B) Unicode removed the original ASCII assignments.
   (C) The first 128 Unicode code points, U+0000–U+007F, use the same single-byte values in UTF-8 as in ASCII.
   (D) ASCII bytes are automatically normalized to NFC.

6. Which operation is best classified as *task-specific normalization* rather than Unicode normalization?
   (A) Converting decomposed `e + combining acute` into precomposed é using NFC
   (B) Reordering combining marks into canonical order
   (C) Lowercasing `Apple` to `apple` because case is irrelevant for the task
   (D) Mapping a Unicode code point to its UTF-8 byte sequence

7. Which statement correctly distinguishes BPE training from BPE encoding?
   (A) Training applies fixed merges; encoding discovers new merges for each input.
   (B) Training learns a vocabulary and ordered merge rules; encoding applies those fixed rules to new text.
   (C) Both training and encoding add every unseen word to the vocabulary.
   (D) Encoding uses test-text pair frequencies to change the learned merge order.

8. A character-level BPE trainer starts from the weighted corpus

   $$2 \times \texttt{new}, \quad 2 \times \texttt{renew}, \quad 1 \times \texttt{set}, \quad 1 \times \texttt{reset}.$$

   After the first merge `n+e` → `ne`, which pair has frequency 4 and is therefore the next merge?
   (A) `r + e`
   (B) `e + t`
   (C) `s + e`
   (D) `ne + w`

9. A learned BPE tokenizer has the ordered merges

   $$1.\ \texttt{n+e} \rightarrow \texttt{ne} \qquad 2.\ \texttt{ne+w} \rightarrow \texttt{new} \qquad 3.\ \texttt{s+t} \rightarrow \texttt{st}.$$

   Starting from characters, how will it encode the unseen word `newest`?
   (A) `[newest]`
   (B) `[new][e][s][t]`
   (C) `[ne][w][e][st]`
   (D) `[new][e][st]`

10. If the number of learned BPE merges k is increased substantially, what general trade-off should we expect?
    (A) Smaller vocabulary and shorter sequences
    (B) Larger vocabulary and usually shorter token sequences
    (C) Larger vocabulary and necessarily longer token sequences
    (D) No effect on either vocabulary size or sequence length

11. In this course, a *document* is best understood as:
    (A) Always one complete physical file on disk
    (B) Always one paragraph
    (C) The unit of text chosen to be represented as one object for the task
    (D) Any text that has already been converted to TF–IDF

12. Suppose the feature vocabulary order is
$$[\texttt{cat}, \texttt{dog}, \texttt{sat}, \texttt{mat}].$$
    What is the Bag-of-Words count vector for `cat sat cat`?
    (A) $[2, 0, 1, 0]$
    (B) $[1, 0, 2, 0]$
    (C) $[2, 1, 0, 0]$
    (D) $[\texttt{cat}, \texttt{sat}, \texttt{cat}, 0]$

13. Using the course convention, what is the term-frequency weight of a term that occurs 10 times in a document?
    (A) 2
    (B) 1
    (C) 10
    (D) $\log_{10}(11)$

14. A corpus has $N = 8$ documents, and the term `the` occurs in all 8. Under the course convention, what is its IDF?
    (A) 8
    (B) 1
    (C) $\log_{10}(8)$
    (D) 0

15. A term occurs 10 times in a document. The corpus has $N = 100$ documents, and the term occurs in 10 documents. What is its TF–IDF weight under the course convention?
    (A) 1
    (B) 2
    (C) 10
    (D) 20

Section B – 2-mark questions

16. Suppose vocabulary size follows Heaps' law
$$|V| = kN^{0.5},$$
    where N is the number of running word instances. A corpus is expanded from N instances to $9N$ instances while remaining in the same statistical regime. Which statement is correct?
    (A) Vocabulary size becomes $9|V|$, and the type–instance ratio is unchanged.
    (B) Vocabulary size is unchanged because a language has a fixed vocabulary.
    (C) Vocabulary size becomes $3|V|$, and the type–instance ratio becomes three times larger.
    (D) Vocabulary size becomes $3|V|$, and the type–instance ratio becomes one-third of its previous value.

17. Which pair of UTF-8 encodings is correct?

U+07FF and U+0800

(A) U+07FF → `DF BF`; U+0800 → `E0 A0 80`
(B) U+07FF → `DF FF`; U+0800 → `E0 80 80`
(C) U+07FF → `C7 BF`; U+0800 → `E0 A0 80`
(D) U+07FF → `07 FF`; U+0800 → `08 00`

18. Consider the following Python code:

```
import unicodedata

a = "\u00E9"       # precomposed e-acute
b = "e\u0301"      # e + combining acute
b_nfc = unicodedata.normalize("NFC", b)
```

Which tuple is correct?

```
(
    len(a), len(a.encode("utf-8")),
    len(b), len(b.encode("utf-8")),
    len(b_nfc), len(b_nfc.encode("utf-8"))
)
```

(A) $(1, 1, 2, 2, 1, 1)$
(B) $(1, 2, 2, 3, 1, 2)$
(C) $(2, 2, 2, 3, 2, 3)$
(D) $(1, 2, 1, 2, 1, 2)$

19. Consider the deterministic pipeline

NFC → lowercase → remove punctuation → fixed BPE → BoW.

The inputs are `US?` and `us!`. At which earliest stage do they become computationally indistinguishable?

(A) NFC normalization
(B) Lowercasing
(C) Fixed BPE encoding
(D) Punctuation removal

20. A BPE trainer uses the ordered weighted corpus

```
[("ac", 1), ("ab", 10)]
```

Ties are resolved by choosing the pair encountered first. The correct first merge is `a+b` → `ab`, but a student's implementation instead chooses `a+c` → `ac`. Which bug most directly explains this result?

(A) The implementation applied merge rules in reverse order.
(B) The implementation performed NFC normalization before counting pairs.
(C) The implementation started from UTF-8 bytes rather than characters.
(D) The implementation counted each distinct word once instead of multiplying pair counts by word frequency.

21. A character-level BPE trainer uses the weighted corpus

$3 \times$ `aba`, $2 \times$ `abb`, $1 \times$ `baa`.

There are no relevant ties. What are the first two merges?

(A) `a+b` → `ab`, followed by `ab+a` → `aba`
(B) `b+a` → `ba`, followed by `a+b` → `ab`
(C) `a+b` → `ab`, followed by `a+a` → `aa`
(D) `b+b` → `bb`, followed by `a+b` → `ab`

22. A BPE trainer processes the single word `aaaa` with frequency 1. When counting, the pair $(\texttt{a}, \texttt{a})$ is counted at every adjacent position. During replacement, non-overlapping occurrences are replaced from left to right. Before the first merge, how many pair occurrences are counted, and how many replacements occur when applying `a+a` $\rightarrow$ `aa`?
   (A) Pair count = 2; replacements = 2
   (B) Pair count = 3; replacements = 3
   (C) Pair count = 3; replacements = 2
   (D) Pair count = 4; replacements = 2

23. A tokenizer has learned the ordered merge rules

    1. `a+b` $\rightarrow$ `ab`
    2. `b+c` $\rightarrow$ `bc`
    3. `ab+c` $\rightarrow$ `abc`
    4. `abc+d` $\rightarrow$ `abcd`.

    What is the final encoding of the unseen word `abbcd`?
   (A) `[ab][bc][d]`
   (B) `[ab][b][c][d]`
   (C) `[a][b][bc][d]`
   (D) `[a][b][b][c][d]`

24. A model accepts at most 4096 token positions. For comparable text, Language E averages 1.25 tokens per word, while Language L averages 3.2 tokens per word. Ignoring boundary effects, approximately how many words fit, and what is the relative capacity?
   (A) E: 3277 words; L: 1280 words; E fits about 2.56 times as many words.
   (B) E: 5120 words; L: 13,107 words; L fits about 2.56 times as many words.
   (C) E: 3277 words; L: 1280 words; both fit the same linguistic content because the token limit is equal.
   (D) E: 1280 words; L: 3277 words; L fits about 2.56 times as many words.

25. A corpus initially has $N = 4$ documents. Term t occurs 5 times in D_1 and appears in no other document. A fifth document is added; it contains t once and introduces a new vocabulary term u. The feature vocabulary is rebuilt from all five documents. What *necessarily* happens to the TF of t in the unchanged document D_1, the IDF of t, and the dimension of D_1's vector?
   (A) TF of t increases, IDF of t increases, and vector dimension stays fixed.
   (B) TF of t is unchanged, IDF decreases from $\log_{10} 4$ to $\log_{10}(5/2)$, and vector dimension increases.
   (C) TF and IDF of t are unchanged because D_1 itself did not change.
   (D) TF of t decreases, IDF becomes zero, and vector dimension decreases.

26. A corpus has $N = 10$ documents. Term t appears in exactly one document. That document is split into two separate documents, and both resulting documents contain t. All other documents remain unchanged. What is the new IDF of t?
   (A) $\log_{10}(10/1) = 1$
   (B) $\log_{10}(11/1) \approx 1.041$
   (C) $\log_{10}(11/2) \approx 0.740$
   (D) $\log_{10}(10/2) \approx 0.699$

27. In a corpus of $N = 1000$ documents, term A occurs 9 times in document d and appears in 100 documents, while term B occurs 2 times in d and appears in 10 documents. Which term has the greater TF–IDF weight in d?
    (A) A, because $9 > 2$
    (B) B, because its larger IDF more than compensates for its smaller term frequency
    (C) They have equal weight
    (D) The comparison cannot be made without knowing the length of d

28. Initially, a corpus has $N = 100$ documents. In document d, term A occurs once with $\mathrm{df}_A = 1$, while term B occurs 10 times with $\mathrm{df}_B = 10$. Then 900 new documents are added. Every new document contains A, and none contains B. Which statement is correct?
    (A) A has greater weight initially and remains greater afterward.
    (B) B has greater weight initially, but A becomes greater afterward.
    (C) The two terms are tied initially, but B has much greater weight after the expansion.
    (D) The two terms remain tied because their counts in d did not change.

29. Initially $N = 100$. In document d, term A occurs 100 times and appears in all 100 documents; term B occurs once in d and appears in 10 documents. Then 900 documents containing neither term are added. Which statement is correct?
    (A) A is larger both before and after the expansion.
    (B) B is larger both before and after the expansion.
    (C) B is larger initially, but A becomes larger after the expansion.
    (D) The weights cannot change because document d did not change.

30. A corpus has $N = 100$ documents. Term x occurs 100 times, all in D_1. Term y occurs once in every document, including D_1. Thus, both terms have total corpus frequency 100. What are their TF–IDF weights in D_1?
    (A) $x = 3,\ y = 1$
    (B) $x = 6,\ y = 2$
    (C) $x = 0,\ y = 6$
    (D) $x = 6,\ y = 0$

End of question paper